Data-Driven Software Engineering: Towards Regulatory Compliance: A few-shot Learning Approach to Extract Processing Activities

Pragyan K C[1] Dr. Mitra Bokaei Hosseini[1]

[1]Computer Science Department, University of Texas at San Antonio

ABSTRACT

The rise of mobile apps relies on user data, making GDPR compliance, like maintaining a Record of Processing Activities (RoPA), essential. Small app developers struggle with this due to limited resources. Our approach uses GPT-3.5 Turbo to generate RoPA segments from user scenarios. Our findings highlight the significant influence of the number of examples in prompts, while demonstrating negligible variability in F1 scores across multiple prompt repetitions. Our prompts achieve successful summarization of processing activities with an average 70% ROUGE-L F1 score. Finally, we discuss avenues for improving results through manual evaluation of the generated summaries.

CONTRIBUTION

(1) a 50-scenario corpus with summarized processing activities;

(2) an empirical evaluation of utilizing the GPT-3.5 Turbo as an instance of LLMs for extractive summarization;

(3) an evaluation of different parameters affecting the LLM instance in extractive summarization task;

(4) a manual evaluation of summaries extracted using the LLM instance.

Find the paper here:

What is the issue?

- GDPR requires the maintenance of a Record of Processing Activities (RoPA) by companies
- Small app-developing companies face challenges in meeting such compliance requirements due to resource limitations and no documentation available.

Records of Processing Act. (RoPA)

User signs up.
App asks name in the registration process.
User tracks calories.
User provides zip code.

Actor | Processing Activities | Data Involved | UI Component

How do we solve this?

Introduce a framework to generate **segments of RoPA** for existing mobile apps **using Large Language Model**

- Framework takes advantage of usage scenarios provided by user

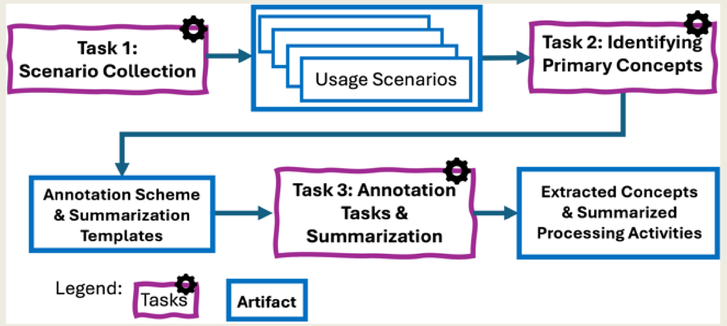
User authored Scenario

I am trying to sign up for this calorie counting app. I am in the registration process and it is asking me for my personal information. I would want to maintain my current healthy weight and track my calories through out the day. I do not want to lose or gain weight through the use of this app. My interactions with this app would be to monitor it as I go through out the day and track how many calories I am consuming each day. The app is asking for my age, it also asks for my country of residence. I am not sure why it is asking for that and my zipcode as well. I feel somewhat uncomfortable providing my zip code because I don't know why the app needs that or what it will do with that information. I might use this app or might not.

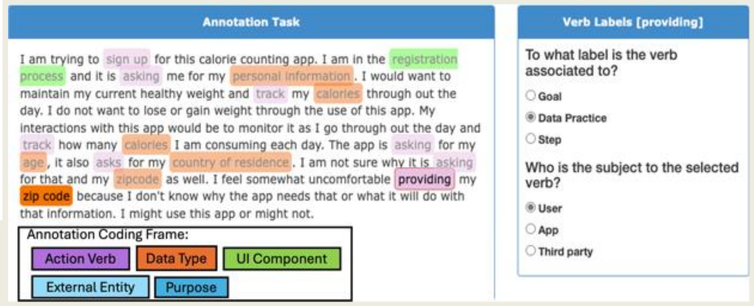
Template for Records of Processing Act.

Goal Template	User <goal-action> <data-type-args> <purpose-args> <external-entity-args>.
Step Template	<actor> <step-action> <UI-args> <purpose-args> <external-entity-args>.
DP Template	<actor> <DP-action> <data-type-args> <UI-args> <purpose-args> <external-entity-args>.

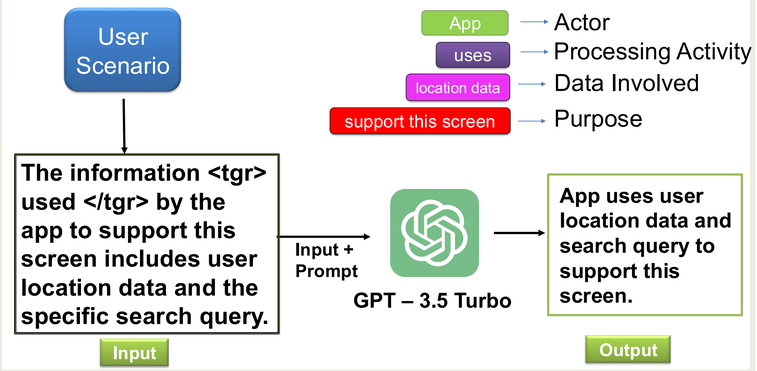
Building the Dataset



Annotation Tool



Input – Output for the Model



Dataset Details

- **50 User Authored Scenarios Collected**

Sentences Extracted		
Goal	**Step**	**Data Practice**
64	**83**	**253**

- Model Used: **GPT 3.5 – Turbo**
- Metrics for Evaluation: **ROUGE-1, ROUGE-2, ROUGE-L, ROUGE-S, METEOR, BERTScore**

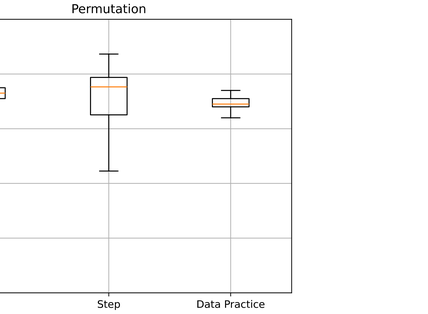
Experiment Design

- Experiment 1: **Find optimal number of shots.**
- Experiment 2: **Permutation of the optimal number.**

Results

Results on Number of Examples

ROUGE-L		
Goal	Step	DP
0.25	0.21	0.22
0.62	0.61	0.53
0.63	0.68	0.54
0.63	0.71	0.57
0.64	0.72	0.55
0.67	0.71	0.60
0.71	0.72	0.63
0.71	0.72	0.63
0.69	0.71	0.63
0.70	0.70	0.62
0.74	0.70	0.63



Results on Order of Examples

Results on Testing Set

Metric	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-S	METEOR	BERTScore
Goal	0.75	0.48	**0.75**	0.36	0.80	0.95
Step	0.75	0.63	**0.76**	0.49	0.85	0.95
Data Practice	0.62	0.42	**0.61**	0.30	0.69	0.94

Manual Analysis Coding Categories	
additional modifiers and adjectives	29/81
incorrect action verb or subject	20/81
missing data type	2/81
missing purpose	6/81
missing UI Component	1/81
summary contains more than two verbs	6/81

Conclusion & Future Works

- Prompted LLM to extract RoPA segments from User Scenarios
- Created a dataset with RoPA from User Scenarios
- Develop NER model for detecting the action verbs
- Use multiple LLMs for more consolidated results
- Evaluate the effectiveness and practical applicability of our framework in extracting RoPA details

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